

Section 12.5: Equations of Lines and Planes

Vectors and the Geometry of Space

MTH 310
Calculus III

Lines in 2D — Review

In 2D, a line is determined by a **point** and a **slope** (direction):

$$y = mx + b \quad \text{or} \quad y - y_0 = m(x - x_0)$$

Example 1

Find the equation of the line through $P(0, 0)$ and $Q(1, 2)$.

Find the slope, then use point-slope form.

$$\text{Slope: } m = \frac{2 - 0}{1 - 0} = 2$$

Using point-slope with $P(0, 0)$: $y - 0 = 2(x - 0)$, so $y = 2x$.

A line needs two things: a **point** and a **direction**.

Parametric Equations of Lines

Instead of $y = mx + b$, we can describe a line **parametrically**:

$$x = x_0 + at, \quad y = y_0 + bt$$

The parameter t sweeps out the line. The **direction vector** $\vec{v} = \langle a, b \rangle$ tells which way the line goes.

Example 2

Write parametric equations for the line through $P(0, 0)$ with direction $\langle 1, 2 \rangle$.

Use the parametric form with the given point and direction.

$$x = 0 + 1 \cdot t = t, \quad y = 0 + 2 \cdot t = 2t$$

Check: when $t = 0$ we get $P(0, 0)$ ✓. When $t = 1$ we get $(1, 2)$ ✓.

Multiple parametrizations: The same line has many parametric descriptions.

Starting at $Q(1, 2)$ and going backward: $x = 1 - s$, $y = 2 - 2s$

Parametric equations are **not unique** — different starting points and direction vectors can describe the same line.

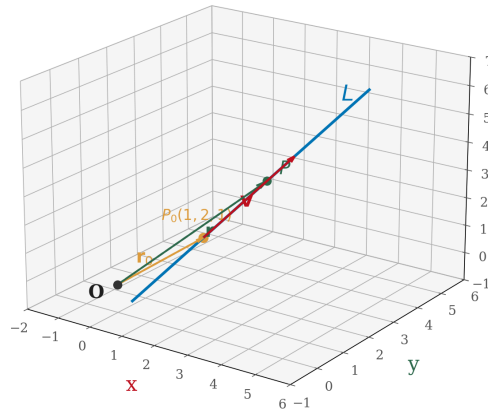
Vector Equation of a Line

Pack the parametric equations into one **vector equation**:

Vector Equation of a Line

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

- $\vec{r}_0$ = position vector of a known point on the line
- $\vec{v}$ = **direction vector** (parallel to the line)
- t = parameter (varies over all real numbers)



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A line in **any dimension** needs just two things: a **point** and a **direction**.

Lines in 3D

The same formula works in 3D — no changes needed!

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

Example 3

Find vector and parametric equations for the line through $P(1, 1, 1)$ and $Q(4, 5, 6)$.

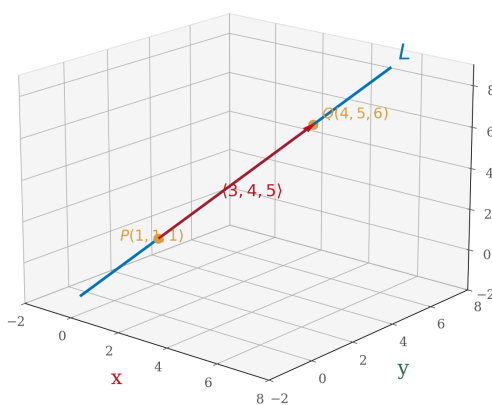
Find the direction vector $\overrightarrow{PQ}$, then write the vector equation. Expand into parametric form.

Direction vector: $\vec{v} = \overrightarrow{PQ} = \langle 4 - 1, 5 - 1, 6 - 1 \rangle = \langle 3, 4, 5 \rangle$

Vector equation: $\vec{r} = \langle 1, 1, 1 \rangle + t\langle 3, 4, 5 \rangle$

Parametric equations:

$$x = 1 + 3t, \quad y = 1 + 4t, \quad z = 1 + 5t$$



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Symmetric Equations

From the parametric equations, **solve each one for t** :

$$t = \frac{x - 1}{3} = \frac{y - 1}{4} = \frac{z - 1}{5}$$

Symmetric Equations

If $\vec{v} = \langle a, b, c \rangle$ (all nonzero), the **symmetric equations** of the line through (x_0, y_0, z_0) are:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Uses:

- Quick way to check if a point is on the line (plug in and see if all three fractions are equal)

- Eliminates the parameter t

If one direction component is **zero**, that coordinate is constant — write it separately. For example, if $a = 0$: $x = x_0$ and

$$\frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Your Turn: Line Through Two Points

Example 4

Find parametric and symmetric equations for the line through $A(2, -1, 3)$ and $B(4, 3, 1)$.

Find the direction vector $\vec{AB}$, write parametric equations, then solve each for t .

Direction vector: $\vec{v} = \overrightarrow{AB} = \langle 4 - 2, 3 - (-1), 1 - 3 \rangle = \langle 2, 4, -2 \rangle$

Parametric equations:

$$x = 2 + 2t, \quad y = -1 + 4t, \quad z = 3 - 2t$$

Symmetric equations:

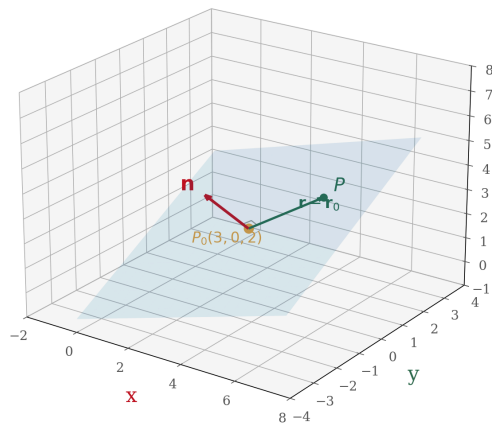
$$\frac{x - 2}{2} = \frac{y + 1}{4} = \frac{z - 3}{-2}$$

Planes in 3D

A **plane** is determined by a point and a **normal vector** (perpendicular to the plane).

If $\vec{r}_0$ points to a known point P_0 and $\vec{n}$ is normal to the plane, then any point $\vec{r}$ in the plane satisfies:

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$$



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Equations of a Plane

- **Vector equation:** $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$
- **Scalar equation:** $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$, where $\vec{n} = \langle a, b, c \rangle$
- **Linear equation:** $ax + by + cz = d$

The coefficients of x , y , z in the plane equation $ax + by + cz = d$ **are** the components of the normal vector:
 $\vec{n} = \langle a, b, c \rangle$.

Example: Equation of a Plane

Example 5

Find the equation of the plane through $P(3, 0, 2)$ with normal vector $\vec{n} = \langle -1, -2, 3 \rangle$.

Use the scalar equation with the given point and normal vector.

Step 1: Use the scalar form $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$:

$$\langle -1, -2, 3 \rangle \cdot \langle x - 3, y - 0, z - 2 \rangle = 0$$

Step 2: Expand the dot product:

$$-(x - 3) - 2(y - 0) + 3(z - 2) = 0$$

Step 3: Simplify:

$$-x + 3 - 2y + 3z - 6 = 0$$

$$-x - 2y + 3z = 3$$

Shortcut: The coefficients of x , y , z come straight from the normal vector. Just plug in the point to find d .

Line-Plane Intersection

Example 6

Where does the line $\vec{r} = \langle 1, 2, -1 \rangle + t\langle 4, -1, 3 \rangle$ intersect the plane $-x - 2y + 3z = 3$?

Substitute the parametric equations into the plane equation and solve for t .

Parametric equations: $x = 1 + 4t$, $y = 2 - t$, $z = -1 + 3t$

Substitute into $-x - 2y + 3z = 3$:

$$-(1 + 4t) - 2(2 - t) + 3(-1 + 3t) = 3$$

$$-1 - 4t - 4 + 2t - 3 + 9t = 3$$

$$7t - 8 = 3$$

$$t = \frac{11}{7}$$

Plug back in:

$$x = 1 + 4\left(\frac{11}{7}\right) = \frac{51}{7}, \quad y = 2 - \frac{11}{7} = \frac{3}{7}, \quad z = -1 + 3\left(\frac{11}{7}\right) = \frac{26}{7}$$

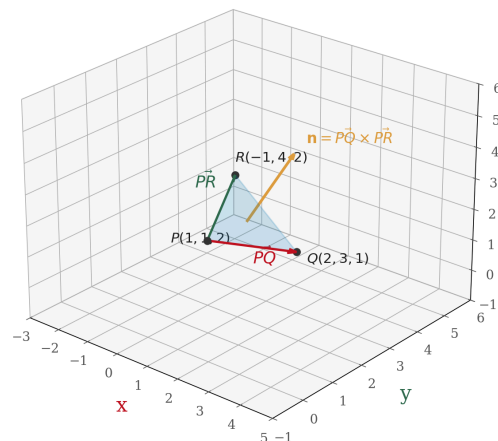
Intersection point: $\left(\frac{51}{7}, \frac{3}{7}, \frac{26}{7}\right)$

Plane Through Three Points

Example 7

Find the equation of the plane through $P(1, 1, 2)$, $Q(2, 3, 1)$, and $R(-1, 4, 2)$.

Strategy: Find two vectors in the plane, then use the **cross product** to get the normal.



Find $\overrightarrow{PQ}$ and $\overrightarrow{PR}$, compute their cross product, then use the result as the normal vector.

Step 1 — Two vectors in the plane:

$$\overrightarrow{PQ} = \langle 2 - 1, 3 - 1, 1 - 2 \rangle = \langle 1, 2, -1 \rangle$$

$$\overrightarrow{PR} = \langle -1 - 1, 4 - 1, 2 - 2 \rangle = \langle -2, 3, 0 \rangle$$

Step 2 — Normal vector $\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$:

$$\vec{n} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -1 \\ -2 & 3 & 0 \end{vmatrix} = \langle (2)(0) - (-1)(3), (-1)(-2) - (1)(0), (1)(3) - (2)(-2) \rangle = \langle 3, 2, 7 \rangle$$

Step 3 — Plane equation using point $P(1, 1, 2)$:

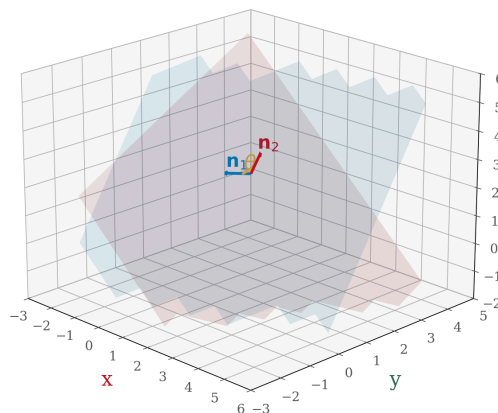
$$3(x - 1) + 2(y - 1) + 7(z - 2) = 0$$

$$3x + 2y + 7z = 19$$

Three points $\rightarrow$ two vectors $\rightarrow$ cross product $\rightarrow$ normal $\rightarrow$ plane equation. This is one of the most important recipes in 12.5.

Angle Between Planes

The **angle between two planes** equals the angle between their normal vectors.



Angle Between Planes

If two planes have normal vectors $\vec{n}_1$ and $\vec{n}_2$, the angle θ between the planes satisfies:

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$$

We use the **absolute value** of the dot product so that $0 \leq \theta \leq 90^\circ$.

Example 8

Find the angle between the planes $x - 2y + z = 2$ and $3x - 2y + 3z = 3$.

Read off the normal vectors, then use the angle formula.

Normal vectors: $\vec{n}_1 = \langle 1, -2, 1 \rangle$, $\vec{n}_2 = \langle 3, -2, 3 \rangle$

$$\vec{n}_1 \cdot \vec{n}_2 = (1)(3) + (-2)(-2) + (1)(3) = 3 + 4 + 3 = 10$$

$$|\vec{n}_1| = \sqrt{1 + 4 + 1} = \sqrt{6}, \quad |\vec{n}_2| = \sqrt{9 + 4 + 9} = \sqrt{22}$$

$$\cos \theta = \frac{10}{\sqrt{6} \sqrt{22}} = \frac{10}{\sqrt{132}}$$

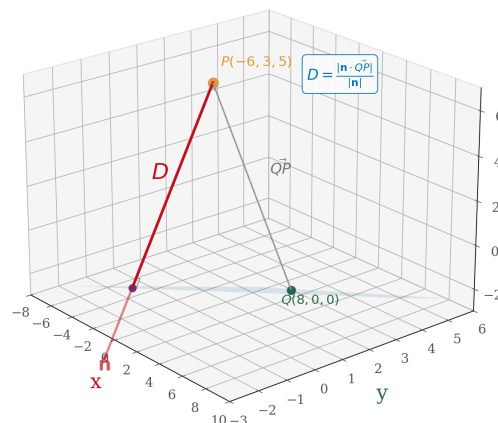
$$\theta = \cos^{-1}\left(\frac{10}{\sqrt{132}}\right) \approx 29.5^\circ$$

Special cases:

- **Parallel planes:** $\vec{n}_1 = c \vec{n}_2$ (normal vectors are scalar multiples)
- **Perpendicular planes:** $\vec{n}_1 \cdot \vec{n}_2 = 0$

Distance from a Point to a Plane

To find the distance from a point to a plane, **project** a connecting vector onto the normal.



Distance from a Point to a Plane

The distance from point $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz = d$ is:

$$D = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Example 9

Find the distance from $P(-6, 3, 5)$ to the plane $x - 2y - 4z = 8$.

Method: pick any point Q on the plane, form PQ , then project onto the normal. Or use the formula.

Method 1 — Projection approach:

Pick $Q = (8, 0, 0)$ on the plane (set $y = z = 0$).

$$\overrightarrow{QP} = \langle -6 - 8, 3 - 0, 5 - 0 \rangle = \langle -14, 3, 5 \rangle$$

Normal: $\vec{n} = \langle 1, -2, -4 \rangle$

$$D = |\text{comp}_{\vec{n}} \overrightarrow{QP}| = \frac{|\vec{n} \cdot \overrightarrow{QP}|}{|\vec{n}|} = \frac{|(-14)(1) + (3)(-2) + (5)(-4)|}{\sqrt{1 + 4 + 16}} = \frac{|-14 - 6 - 20|}{\sqrt{21}} = \frac{40}{\sqrt{21}}$$

Method 2 — Formula:

$$D = \frac{|(-6)(1) + (3)(-2) + (5)(-4) - 8|}{\sqrt{1 + 4 + 16}} = \frac{|-6 - 6 - 20 - 8|}{\sqrt{21}} = \frac{40}{\sqrt{21}}$$

Geometry True/False — Lines and Planes in 3D

Let's test our spatial intuition! Decide whether each statement is **True** or **False**.

Statement	T/F
Two planes either intersect or are parallel.	
Two lines either intersect or are parallel.	
Two planes parallel to a third plane are parallel.	
Two planes orthogonal to a third plane are parallel.	
Two lines parallel to a plane are parallel.	
Two lines orthogonal to a plane are parallel.	
A plane and a line either intersect or are parallel.	

Think carefully about each one. Sketch examples and counterexamples.

| Statement | T/F | Reason |

| — | — | — |

| Two planes either intersect or are parallel. | **T** | Planes are infinite — if they aren't parallel, they must cross somewhere. |

| Two lines either intersect or are parallel. | **F** | In 3D, lines can be **skew** (not parallel and not intersecting)! |

| Two planes parallel to a third plane are parallel. | **T** | All three share the same normal direction. |

| Two planes orthogonal to a third plane are parallel. | **F** | Think of two walls meeting at a corner — both perpendicular to the floor, but not parallel to each other. |

| Two lines parallel to a plane are parallel. | **F** | They can point in different directions within (or parallel to) the plane. |

| Two lines orthogonal to a plane are parallel. | **T** | Both must be parallel to the normal vector $\vec{n}$. |

| A plane and a line either intersect or are parallel. | **T** | (Or the line lies in the plane, a special case of "parallel.") |

In 3D, lines have a **third option: skew** — not parallel and not intersecting. This never happens in 2D!

Homework

Section 12.5: 2, 9, 10, 16, 21, 25, 30, 33, 57, 64, 71